GUNEYKAN OZGUL

New York City, NY, USA
guneykanozgul@gmail.com

EDUCATION

Pennsylvania State University 2021-2025
Ph.D. in Computer Science & Engineering *State College, PA*

Dissertation: Quantum Algorithms For Markov Chain Methods In Machine Learning and Optimization
Overall GPA: 3.88

Boston University 2018-2019
Graduate Student in Physics *Boston, MA*

Theoretical Particle Physics and Cosmology
Changed program, no degree obtained

Bogazici University 2012-2018
B.S. in Computer Engineering *Istanbul, TR*

Overall GPA: 3.79 (High honor, Ranked 2nd in graduating class)
Senior Project: A Hybrid Recommendation System with Application on Sparse Twitter Data

Bogazici University 2012-2018
B.S. in Physics (Double Major) *Istanbul, TR*

PUBLICATIONS, PATENTS AND PREPRINTS

1. Augustino B., Herman D., **Ozgul, G**, Watkins, J., Acharya A., Fontana E., Kim J.L., Chakrabarti S. Quantum speedups for group relaxations of integer linear programs. Preprint.

2. Herman D., **Ozgul, G**, Apte A., Kim J.L., Prakash A., Shen J., Chakrabarti S. Mechanisms for quantum advantage in global optimization of nonconvex functions. 29th Annual Quantum Information Processing Conference (QIP 2026).

3. **Ozgul, G.**, Li, X., Mahdavi, M., and Wang, C. Quantum speedups for sampling and non-convex Optimization with Stochastic Oracles. 21st International Conference on Theory of Quantum Computation, Communication and Cryptography (TQC 2026).

4. **Ozgul, G.**, Li, X., Mahdavi, M., and Wang, C. Quantum speedups for monte carlo markov chain methods with application to optimization, 2025. Preprint.

5. Chakrabarti, S., Herman, D., **Ozgul, G.**, Zhu, S., Augustino, B., Hao, T., He, Z., Shaydulin, R., and Pistoia, M. Generalized short path algorithms: Towards super-quadratic speedup over markov chain search for combinatorial optimization. 20th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2025).

6. **Ozgul G.**, Li X., Mahdavi M., and Wang C. Stochastic quantum sampling for non-logconcave distributions and estimating partition functions. Proceedings of the 41st International Conference on Machine Learning, volume 235 of Proceedings of Machine Learning Research, pp. 38953–38982.

7. Akpinar I.E., Akpinar M.C., Tunca C., **Ozgul G**, Isik S., Akhan M.B. Methods and systems of reducing the spread of contagions. US Patent 11,356,810.

WORK EXPERIENCE

JPMorgan & Chase 2025 - Present
Applied Research Associate Sr *New York City, NY*

JPMorgan & Chase Summer 2024
Applied Research Associate Sr Intern *New York City, NY*

The Pennsylvania State University 2023-2025
Graduate Research Assistant *State College, PA*

Pointr Deep Location Company 2019-2021
Graduate Teaching Assistant *Istanbul, TR*

TEACHING EXPERIENCE

Pennsylvania State University 2021-2023
Graduate Teaching Assistant *State College, PA*

Fall 2021 - CMPSC 497 (Introduction to Quantum Computing)
Spring 2022 - CMPSC 448 (Machine Learning and Algorithmic AI)
Fall 2022 - CMPSC 497 (Introduction to Quantum Computing)
Spring 2023 - CMPSC 448 (Machine Learning and Algorithmic AI)

Boston University 2018-2019
Graduate Teaching Assistant *Boston, MA*

Fall 2018 - PY 211 (Intermediate Electromagnetism)
Spring 2019 - PY 106 (Elementary Electromagnetism)
Summer 2019-1 - PY 212 (Intermediate Mechanics)
Summer 2019-2 - PY 211 (Intermediate Electromagnetism)

OTHER PROJECTS AND ROLES

Inzva, AI Projects 4 2020
Project Lead *Istanbul, TR*

Text Classification using Graph Neural Networks

Huawei 2017
Trainee on 5G Technology *Beijing, China*

Selected by Huawei training in Beijing and Shenzen

Bogazici University 2017
Undergraduate Researcher *Istanbul, TR*

Variance Reduction in Multidimensional Monte Carlo Integration
Under supervision of Prof. Taylan Akdogan

TECHNICAL SKILLS

Languages	Turkish (Native), English (Near Native), Spanish (Beginner)
Computer Languages	(Advanced) C/C++, Python, JAVA, Matlab, Mathematica
Tools	Git, LATEX
Frameworks	Qt, Unity, Tensorflow, Pytorch